

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/FINAL/F_DEB_ALKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3330, C2 = 6646, C3 = 1212)

text_priors <- '
Distrib (Fm,          TruncNormal_cv,          0.1,          0.2,          0.0001,          0.8);
Distrib (pAm,         TruncNormal_cv,          100,          0.2,          40,          2000);
Distrib (r_pAm_pM,    TruncNormal_cv,          0.8,          0.1,          0.1,          2);
Distrib (v,           TruncNormal_cv,          0.18505,        0.1,          0.001,          2);
Distrib (kapJ,        Uniform,                  0,              1);
Distrib (kapA,        Uniform,                  0,              1);
Distrib (Ehp,         TruncNormal_cv,          100,          0.2,          20,          300);
Distrib (E_coc,       TruncNormal_cv,          30,          0.2,          1,          300);
Distrib (rOM_ClxHorse, LogUniform,            0.00001,        0.3);
Distrib (K,           Uniform,                  0.00001,        2);
Distrib (dv,          TruncNormal_cv,          2,            0.5,          0,          10);
Distrib (de,          LogUniform,            0.0000000001,    0.1);
Distrib (tau,         TruncNormal_cv,          7,            0.5,          0,          50);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kapJ.1.	kapA.1.
Result_mode	0.005430240	158.4770	0.67427200	0.19333400	0.90895000	0.4781260
2.5%	0.003350630	144.9950	0.57840605	0.15418000	0.86476495	0.3958020
97.5%	0.009074840	185.4000	0.80559480	0.22479300	0.94248055	0.5352729
Min	0.002569130	64.2106	0.50705500	0.12059500	0.44720900	0.0258501
Max	0.207477000	201.4340	0.87108700	0.26514300	0.99191000	0.5753810
Result_mean	0.005707811	164.6265	0.69943118	0.18806159	0.90472974	0.4684237
Result_sd	0.002073058	10.7449	0.05471631	0.01812461	0.01997393	0.0366190

	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.
Result_mode	110.32700	29.756100	0.006115610	0.03576120	3.9409000
2.5%	69.51349	22.910800	0.004843650	0.02136266	3.4276600
97.5%	139.36100	35.876200	0.009635730	0.06045331	5.7443700
Min	48.65130	19.233400	0.000278698	0.01465990	1.6057800
Max	167.07900	46.993300	0.038497800	1.99564000	6.4758100
Result_mean	105.25146	28.798786	0.007193460	0.03708394	4.4896085
Result_sd	17.54027	3.290182	0.001236911	0.01670054	0.6014475

	de.1.	tau.1.	Sigma_W.1.
Result_mode	1.816980e-09	17.791400	0.11030500
2.5%	1.816980e-09	9.225180	0.09814354
97.5%	1.137980e-05	22.209715	0.12662300
Min	1.556330e-09	0.924651	0.08835930
Max	1.351770e-05	28.645900	5.64717000
Result_mean	4.994265e-06	15.921599	0.11167160
Result_sd	4.326172e-06	3.290741	0.05155249

Number of observations, n = 277
Number of parameters, k = 13
Log-likelihood, LL = -9.607968
BIC = 92.32816

Potential scale reduction factors:

	Point est.	Upper C.I.
de.1.	1.01	1.03
dv.1.	1.10	1.27
E_coc.1.	1.03	1.07
Ehp.1.	1.09	1.26
Fm.1.	1.04	1.12
K.1.	1.02	1.05

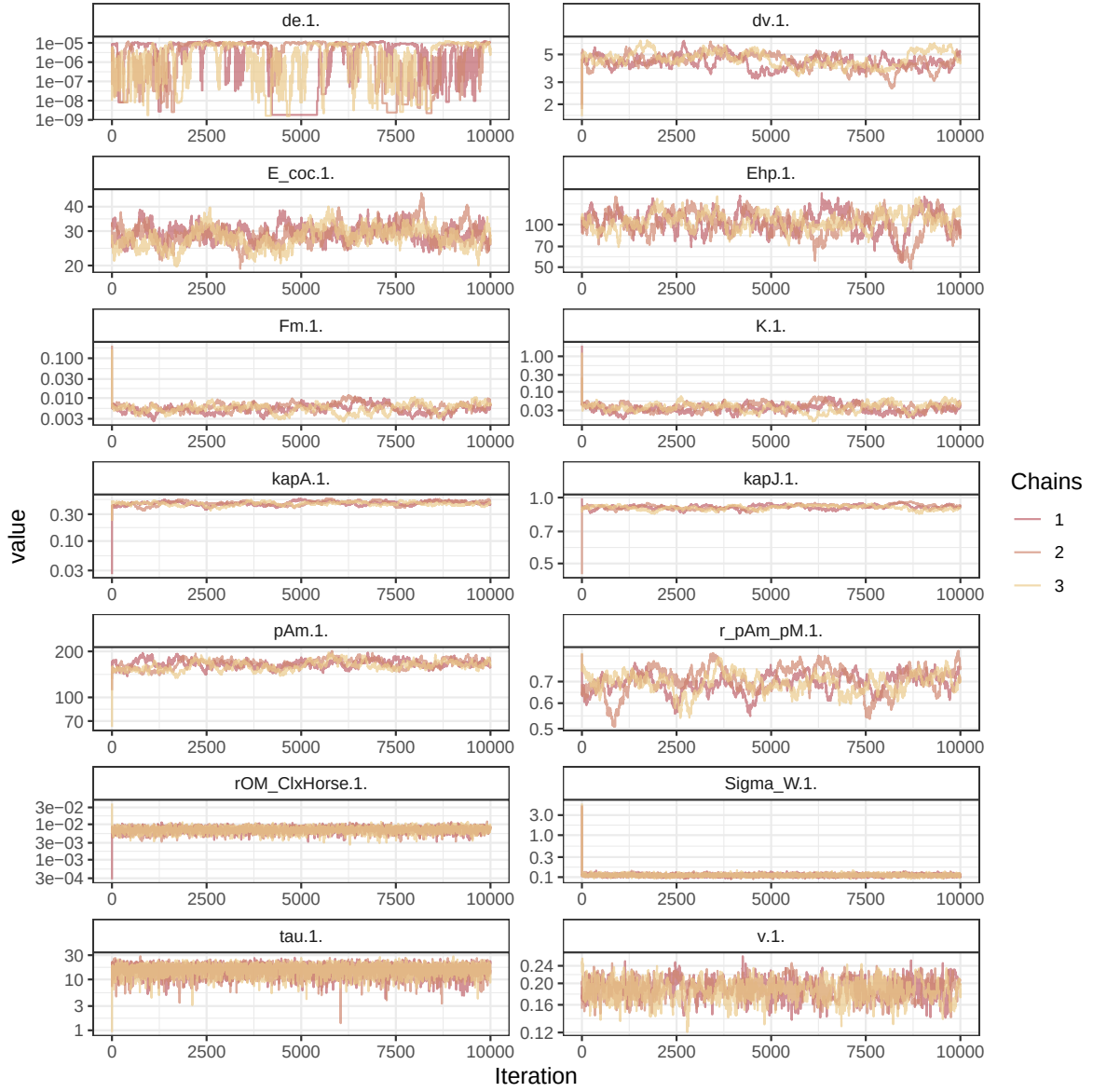


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

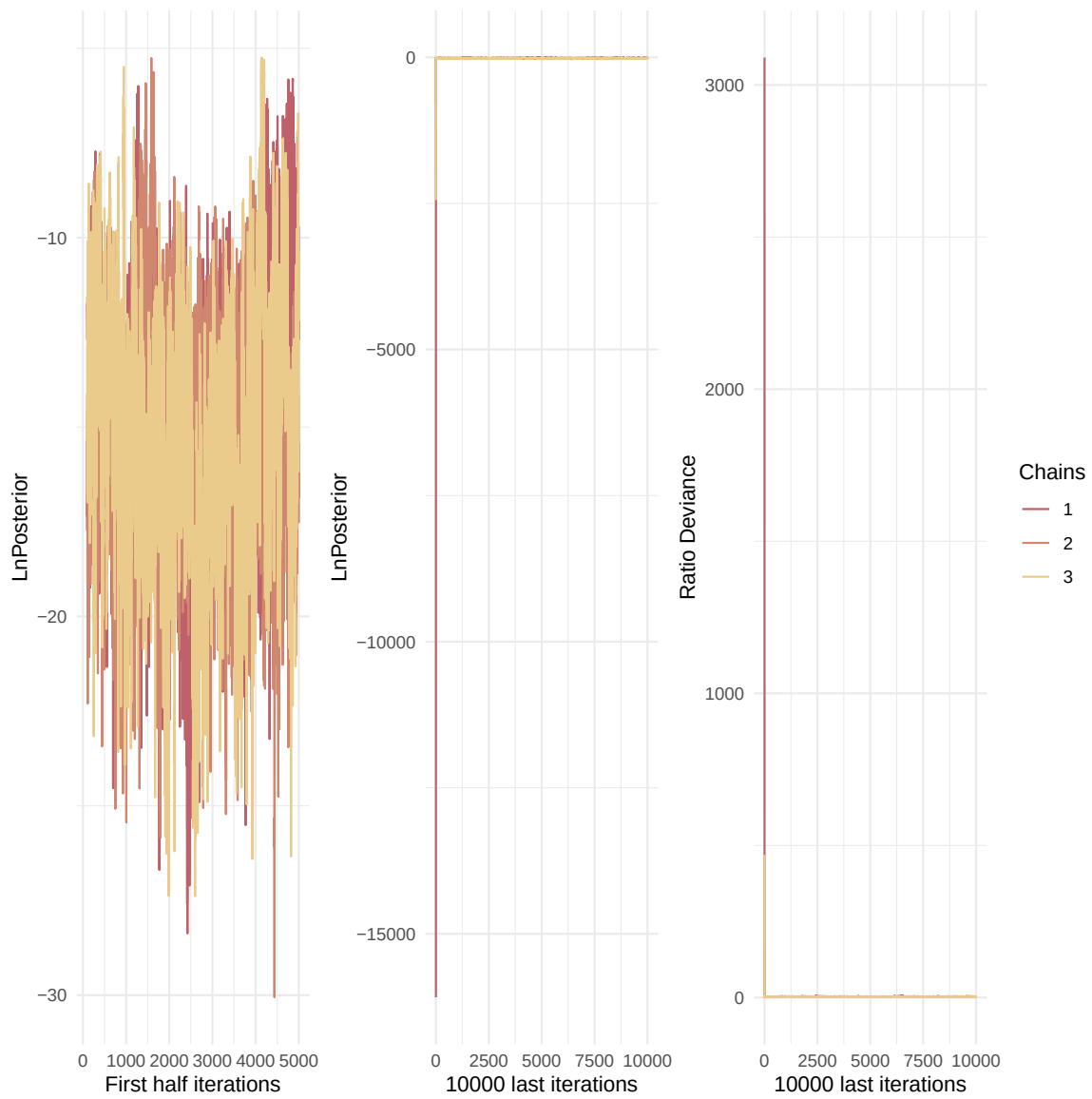


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

kapA.1.	1.19	1.56
kapJ.1.	1.19	1.55
pAm.1.	1.07	1.22
r_pAm_pM.1.	1.12	1.33
rOM_ClxHorse.1.	1.00	1.01
Sigma_W.1.	1.00	1.00
tau.1.	1.00	1.01
v.1.	1.01	1.03

Multivariate psrf

1.23

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

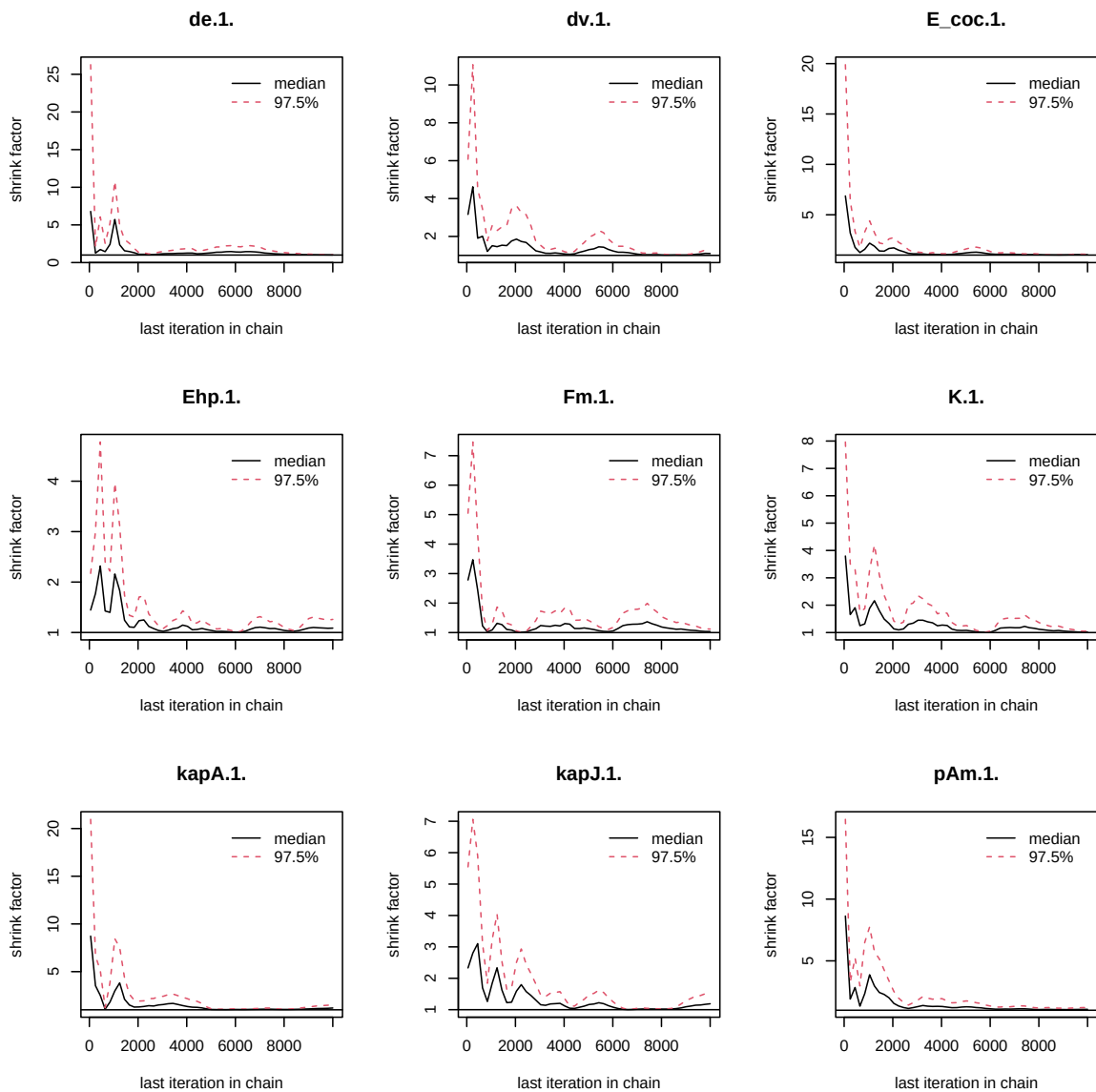


Figure 3: Chains convergence.

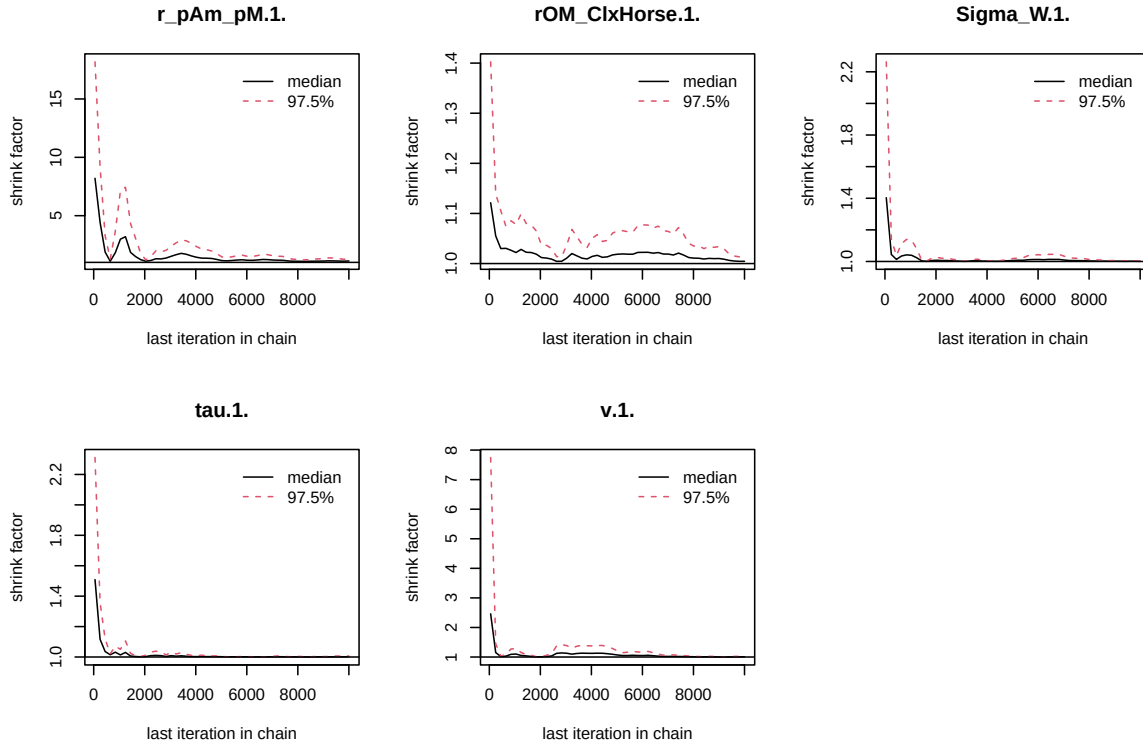


Figure 4: Chains convergence.

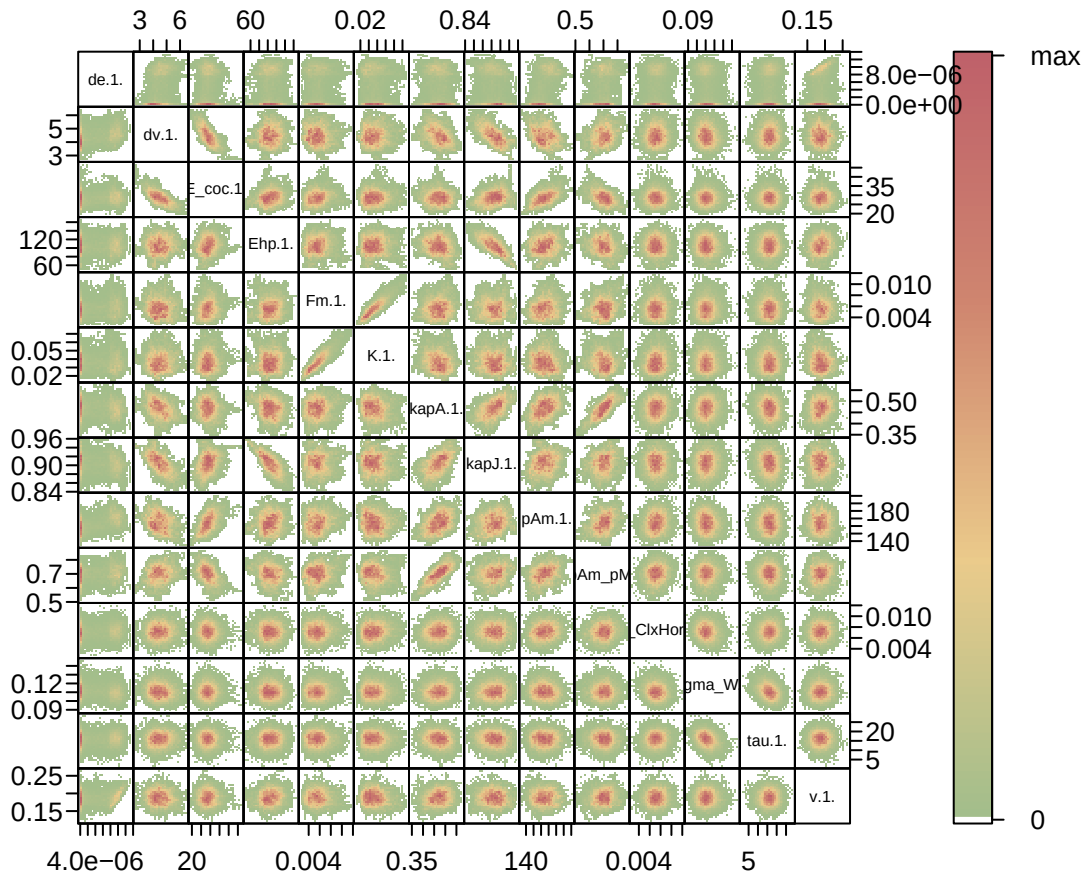


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

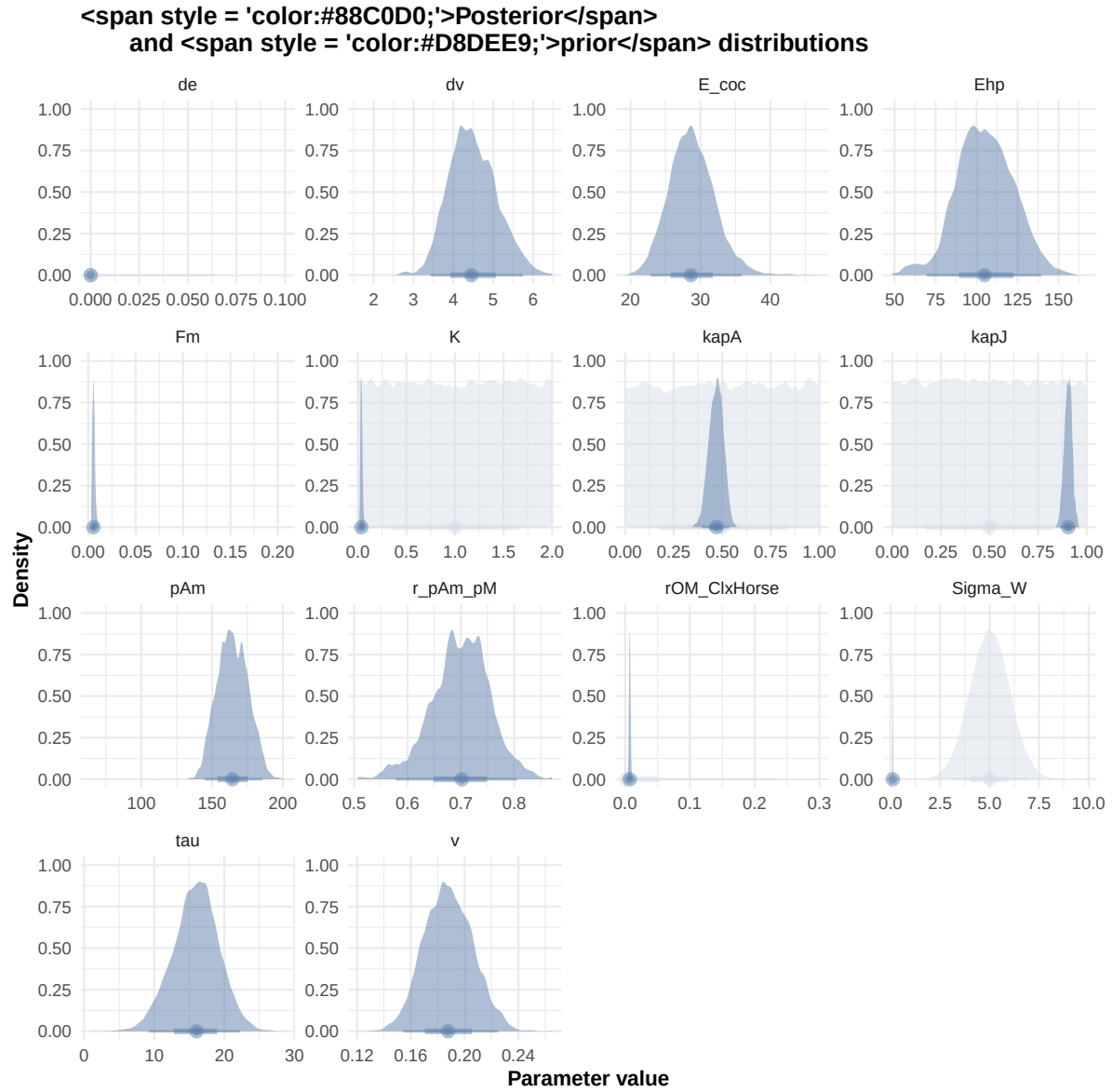


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

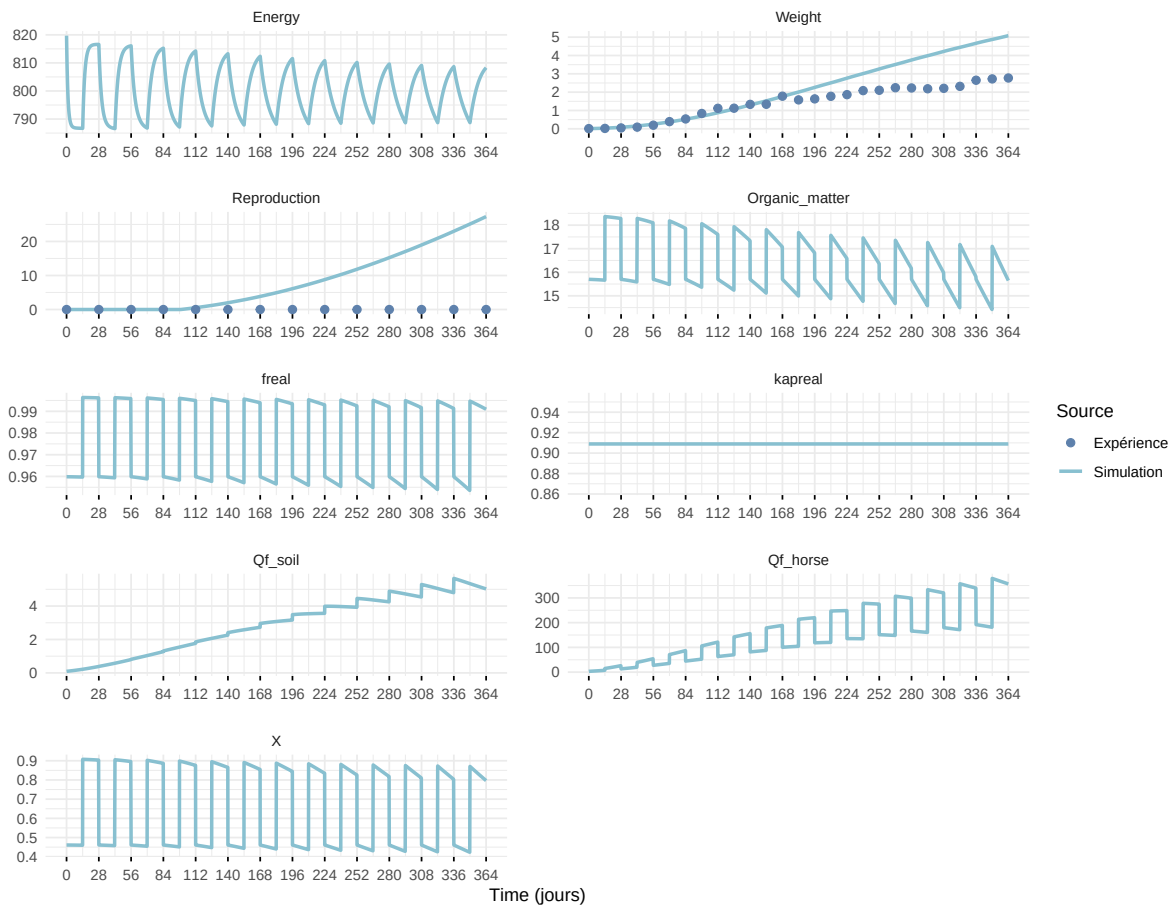
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected join
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship =`

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



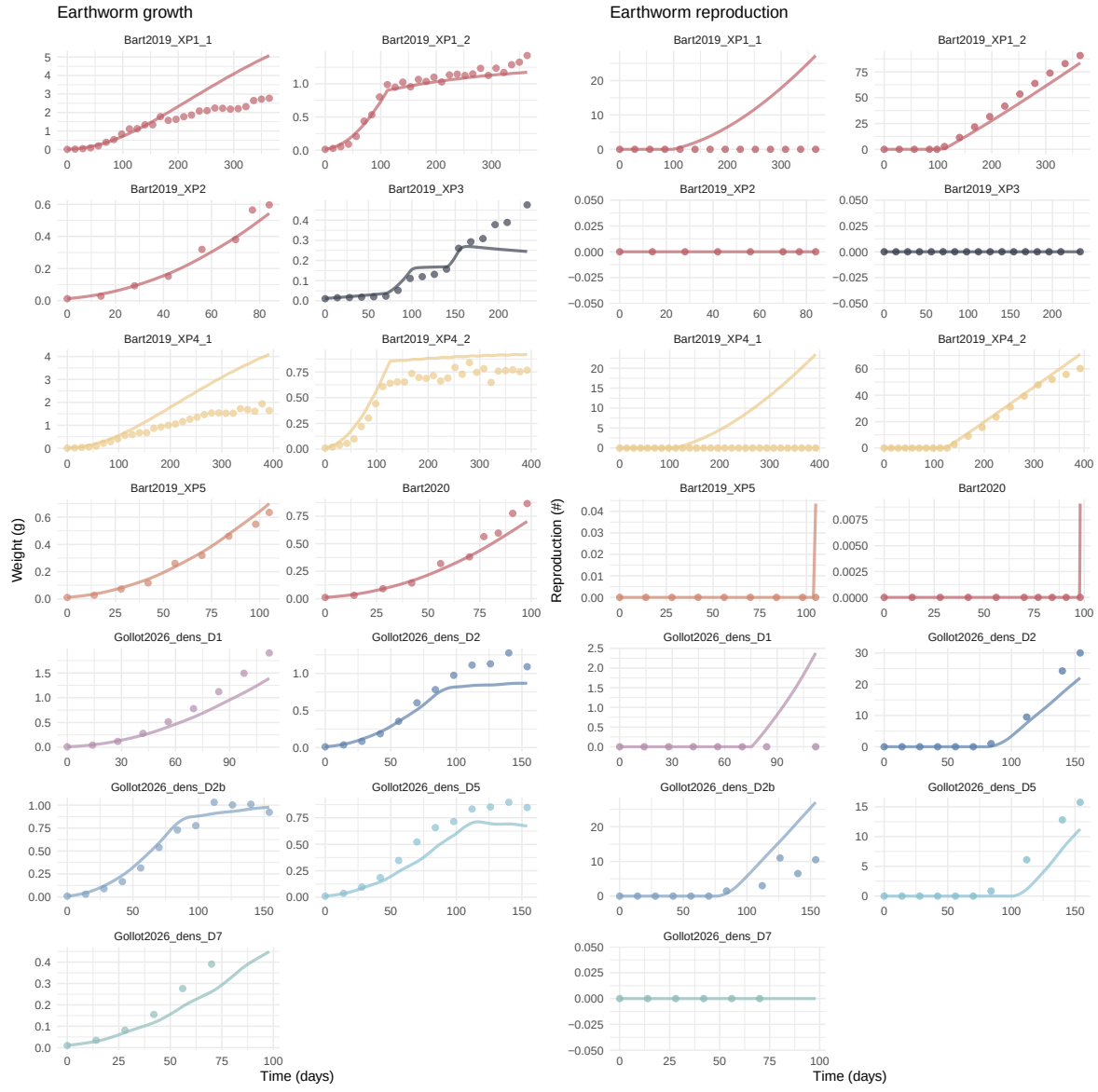
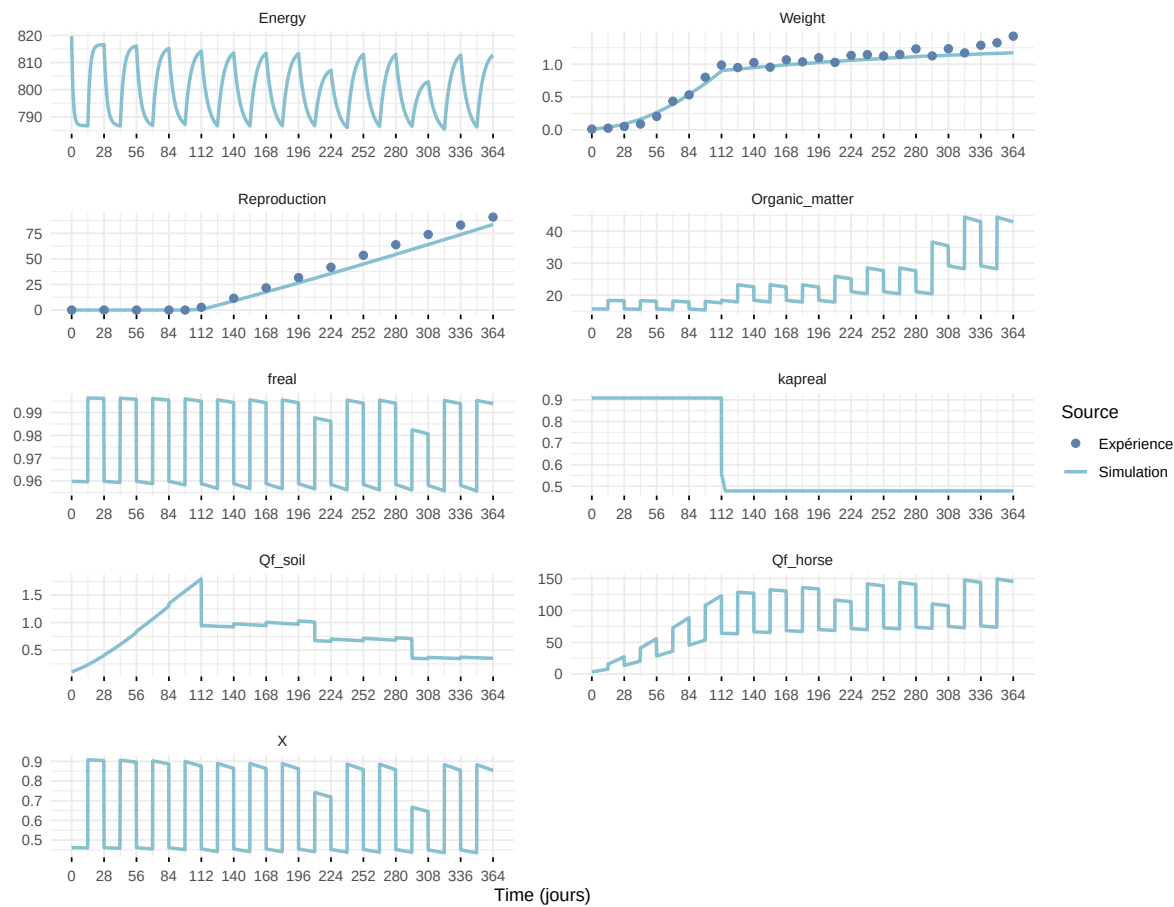
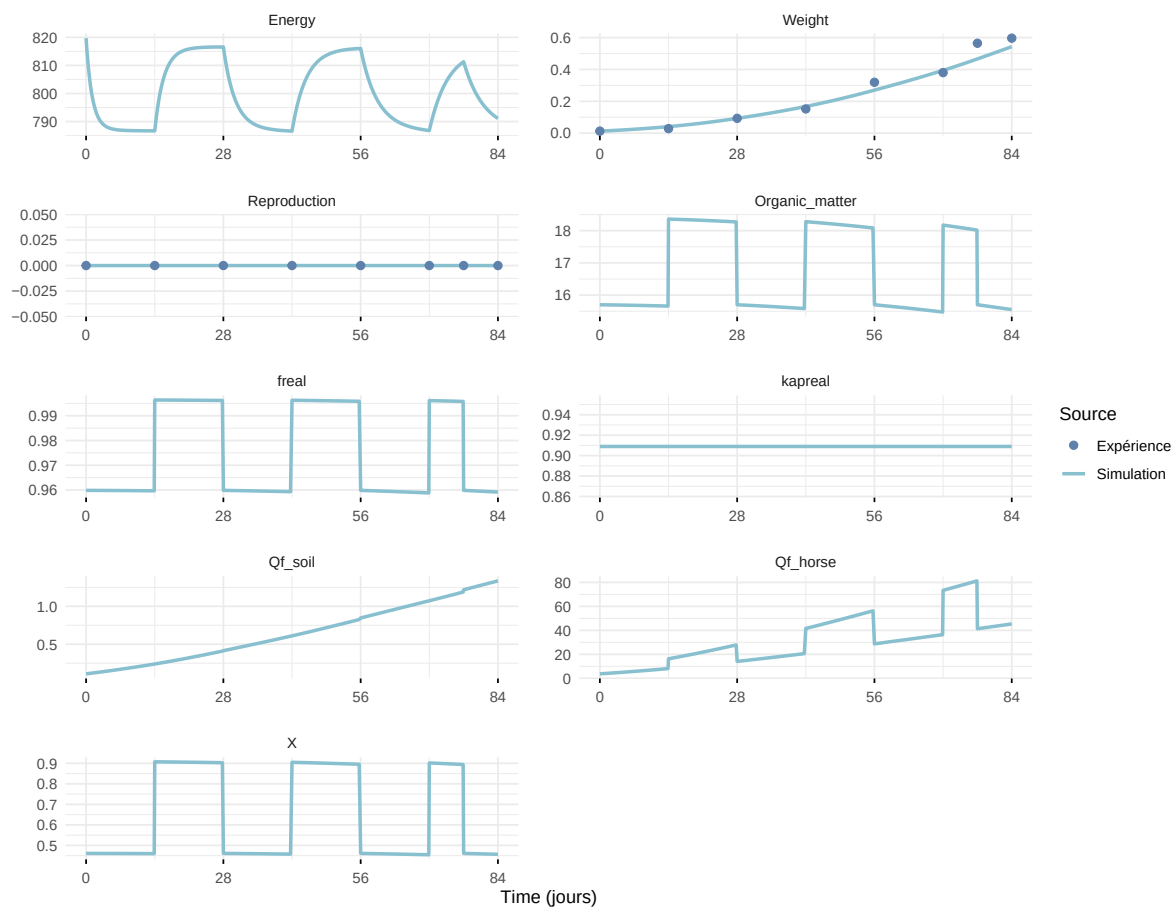


Figure 7: Simulations Weight and Reproduction.

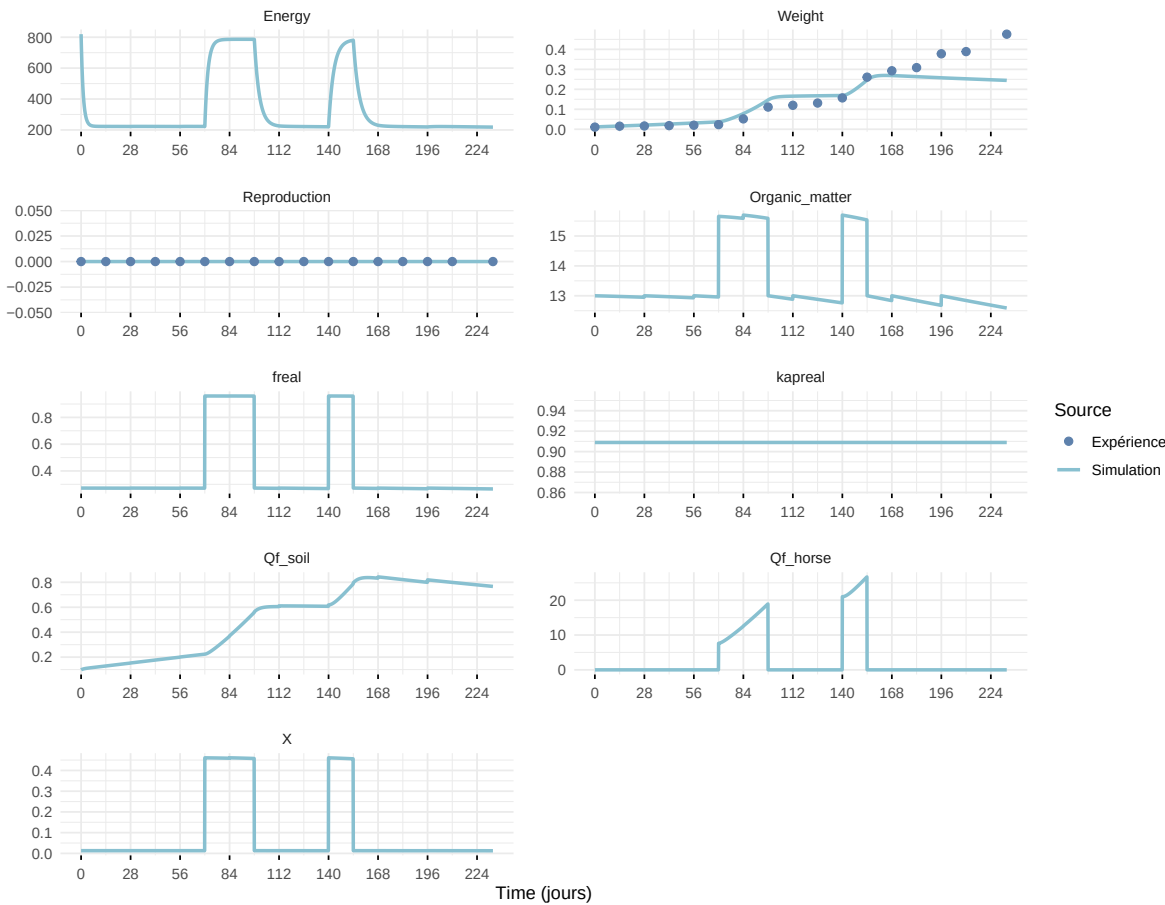
Expérience : Bart2019_XP1_2



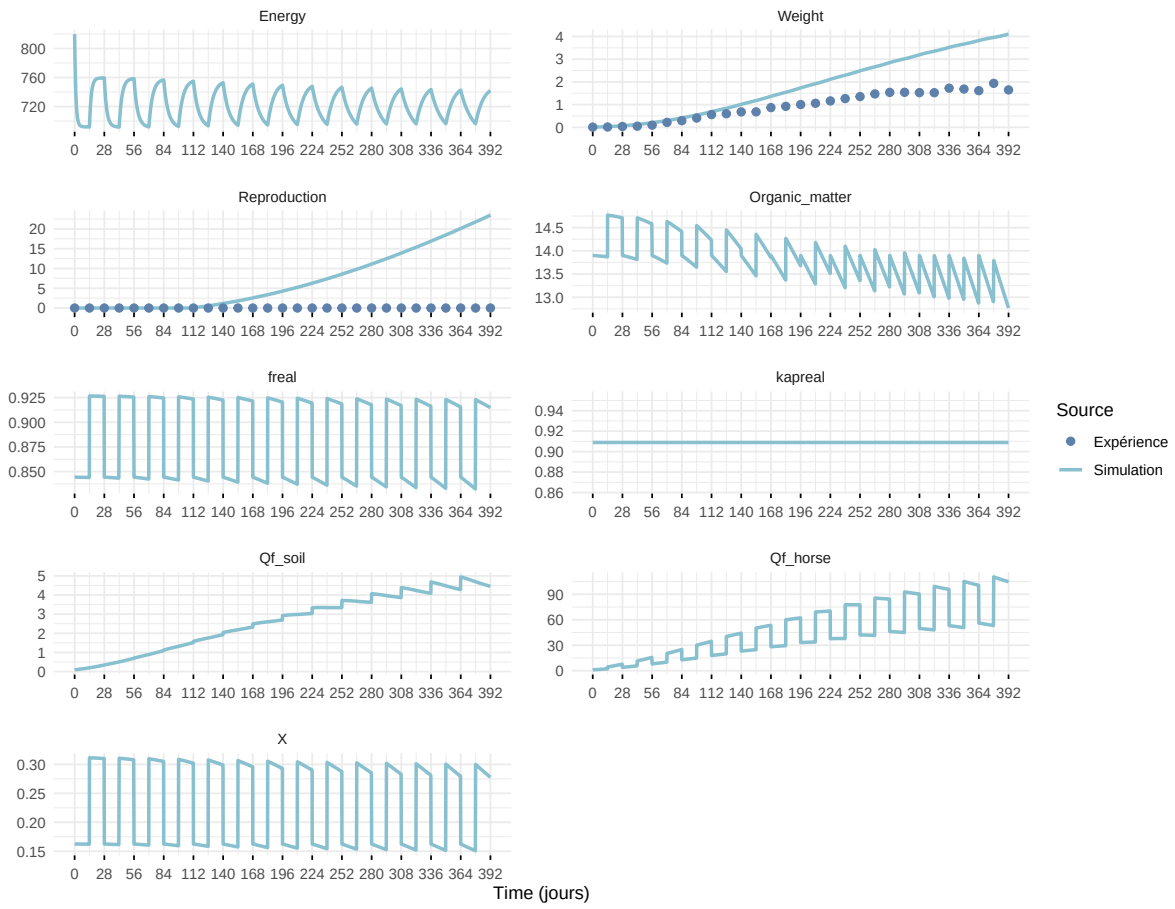
Expérience : Bart2019_XP2



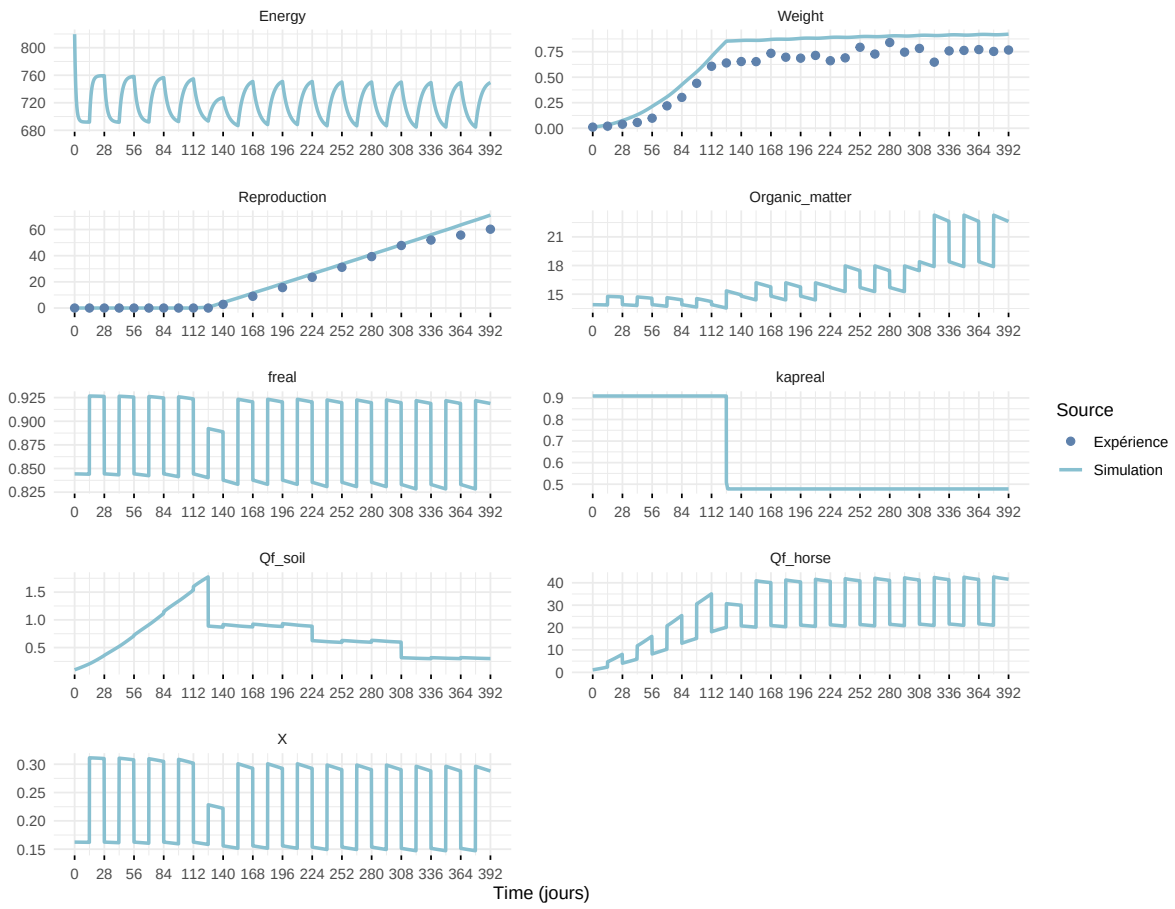
Expérience : Bart2019_XP3



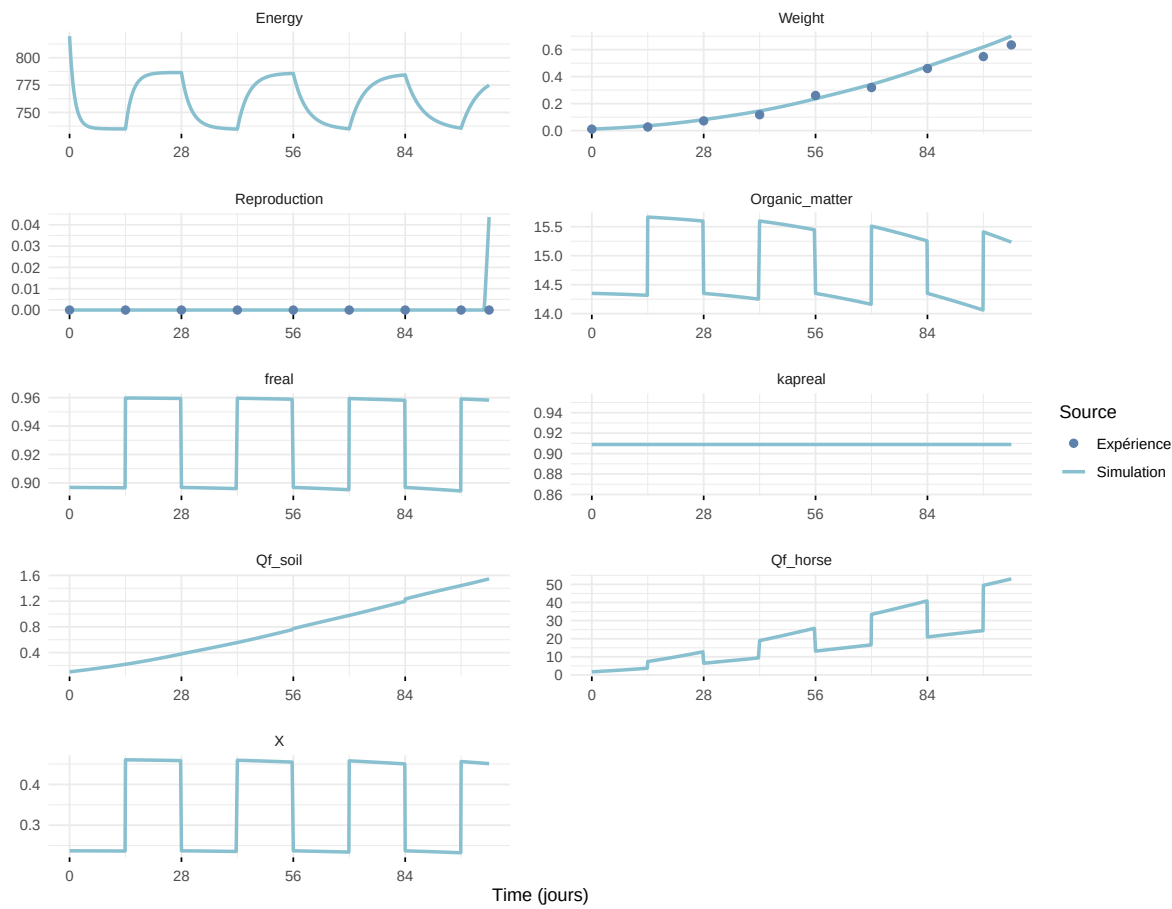
Expérience : Bart2019_XP4_1



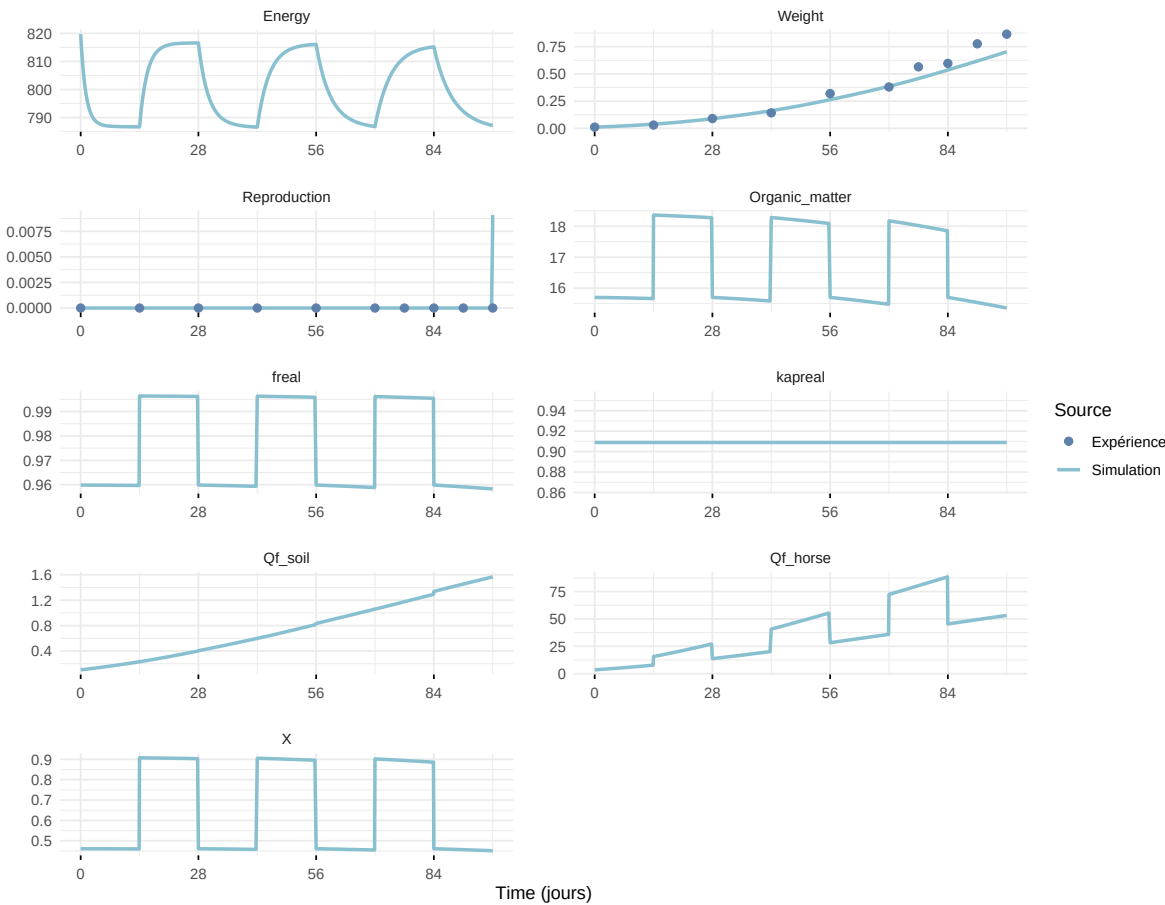
Expérience : Bart2019_XP4_2



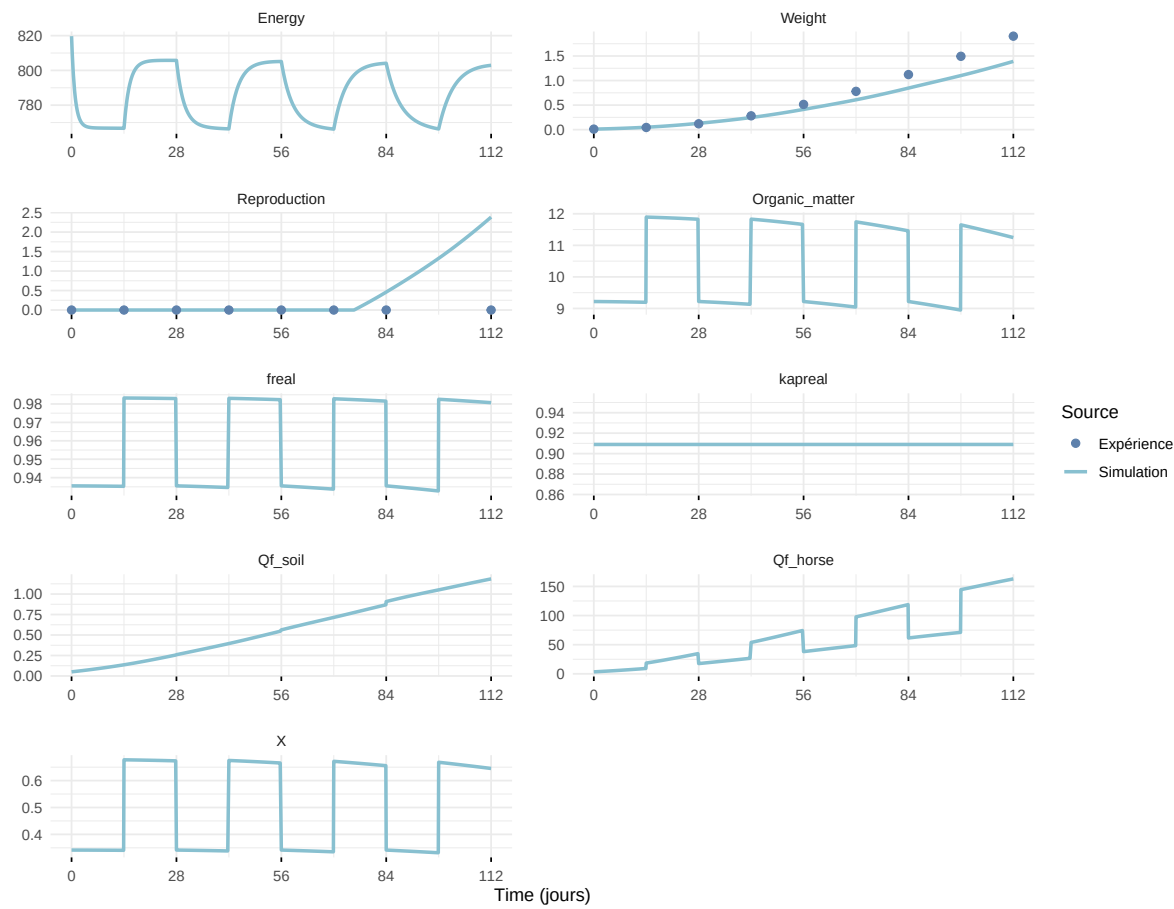
Expérience : Bart2019_XP5



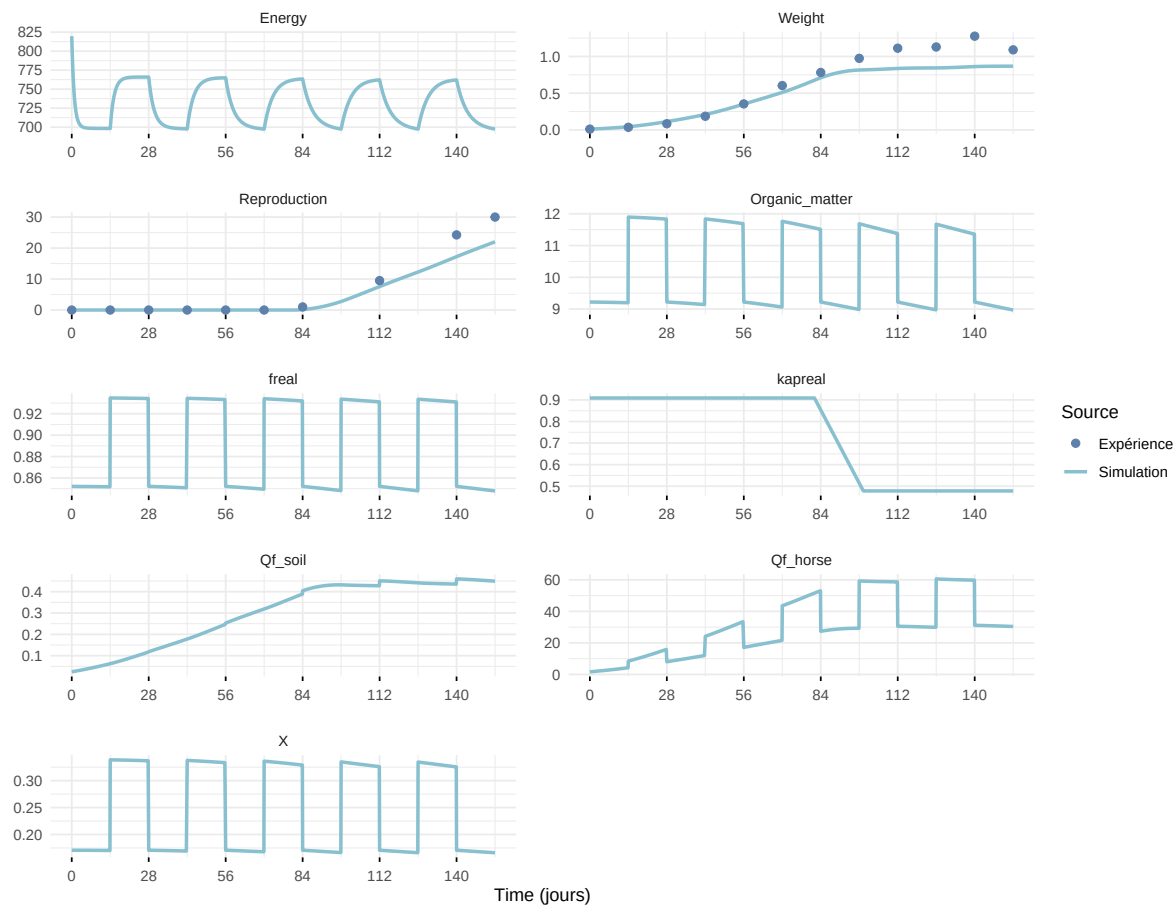
Expérience : Bart2020



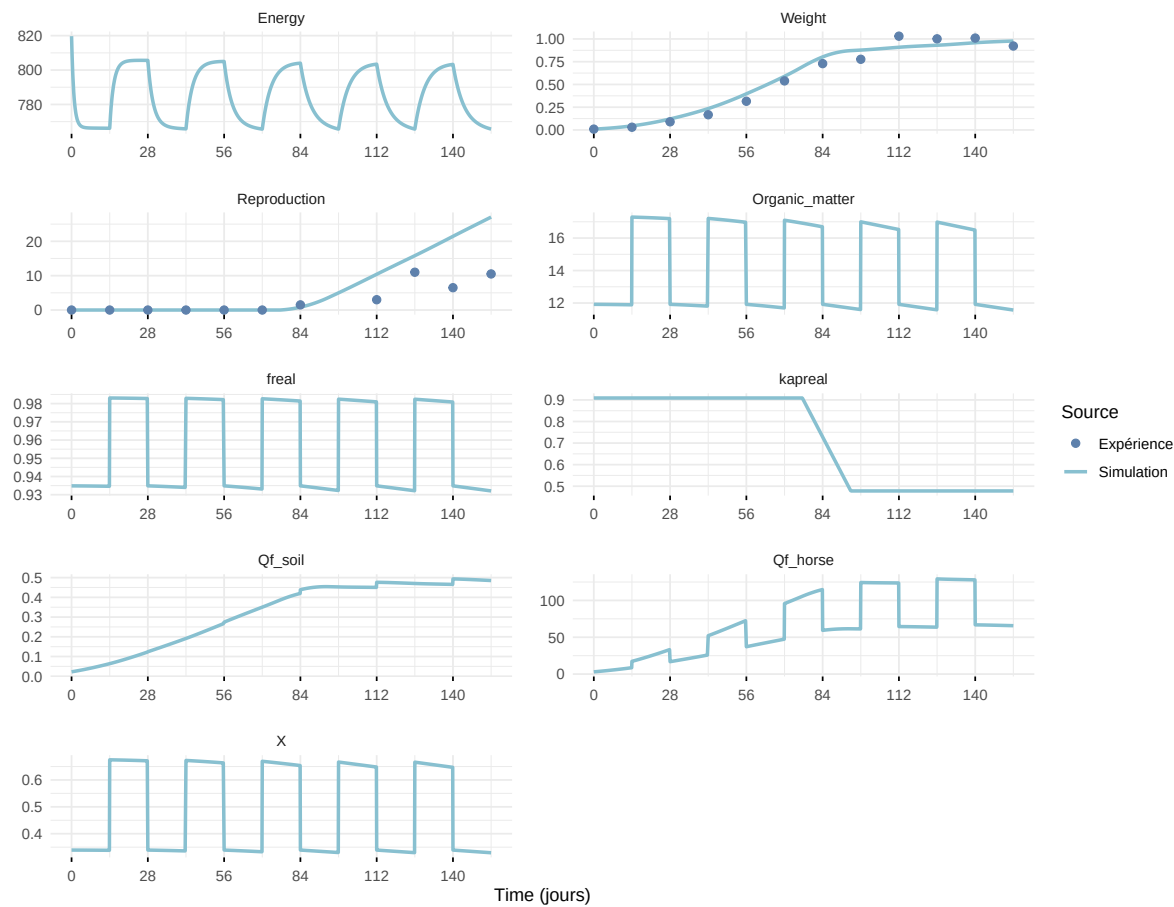
Expérience : Gollot2026_dens_D1



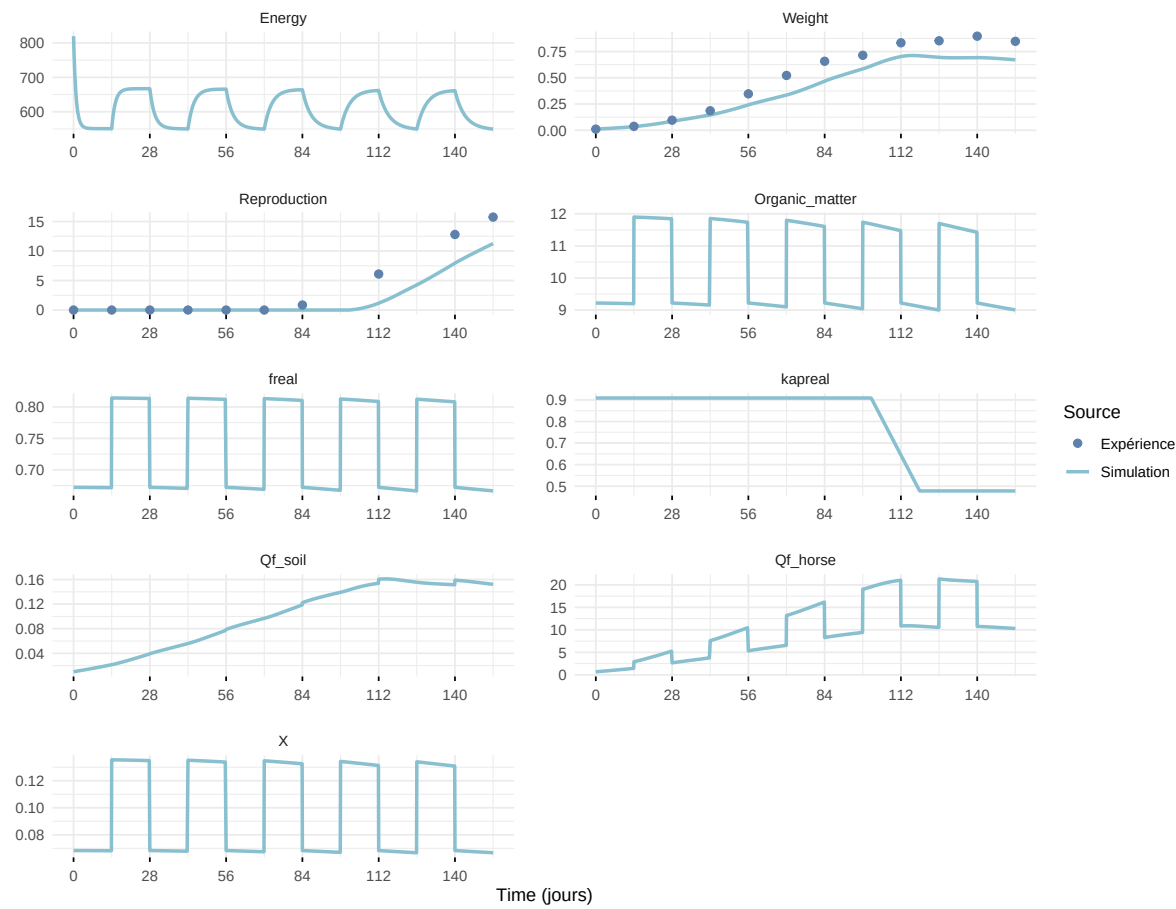
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

